KNECK OFFICIAL

Chemistry — Paper 3

233/3 • ORIGINAL REVISION SET 01 • 2 hours 15 minutes • 40 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Teacher-supervised practical practice. Wear goggles and a coat; no tasting, direct smelling or mouth pipetting. Use dilute reagents only. A supplied-data option is provided when no laboratory is available; identify it as simulated work, not an actual experiment.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SUPPLIED MATERIAL / RESOURCE

Teacher preparation per pair: burette, pipette/filler, conical flasks, white tile; ~150 cm³ 0.100 M HCl, ~100 cm³ approximately 0.050 M Na2CO3, methyl orange; dilute CuSO4, NaOH, aqueous ammonia, BaCl2 and dilute HCl for microscale tests. Collect copper/barium waste separately. Rate task may use supplied data only.

PRACTICAL TASKS

Attempt all three tasks.

1. Titration: solution A is 0.100 mol/dm³ HCl. Solution B is Na2CO3 of unknown concentration. Pipette 25.0 cm³ B into a flask, add methyl orange and titrate with A. Repeat to obtain two titres within 0.20 cm³. (a) Record initial, final and titre for three runs, to 0.05 cm³, and calculate concordant mean. (6) (b) State endpoint change. (2) (c) Write equation and calculate B concentration and mass concentration. (6) (d) State two precautions. (2) For simulated work use: initial/final cm³: 0.00/25.10; 0.20/25.20; 0.00/25.00. [16 marks]

2. Qualitative analysis: the teacher supplies a copper(II) sulfate solution C. (a) Add NaOH dropwise and then in excess. Record observation and inference. (4) (b) To a fresh portion add aqueous ammonia dropwise then in excess. Record observation and inference. (4) (c) Acidify a fresh portion with dilute HCl and add barium chloride. Record result and ionic equation. (4) If using supplied observations instead of reagents, predict each observation and label your work as a simulation. [12 marks]

3. Rate experiment: teacher demonstrates reaction between dilute sodium thiosulfate and dilute HCl over a cross, in a fume cupboard; sulfur dioxide must not be inhaled. Supplied results: thiosulfate volume 50,40,30,20 cm³; water 0,10,20,30 cm³; acid 10 cm³ each; times 25,31,42,63 s. (a) Calculate 1/t for each. (4) (b) Plot 1/t against thiosulfate volume. (3) (c) Explain trend using collision theory. (3) (d) Give two controlled variables. (2) [12 marks]

END OF QUESTIONS

MARKING GUIDANCE

Chemistry • Paper 3 • Set 01

Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

PRACTICAL TASKS

1. (16 marks) Simulated titres 25.10,25.00,25.00; mean acceptable 25.00 (last two) or 25.03 using all concordant titres (6). Yellow to orange at endpoint (2). $Na2CO3+2HCl \rightarrow 2NaCl+CO2+H2O$; nHCl=.0025; ncarbonate=.00125; c=.0500 mol/dm³; Mr=106; mass concentration=5.30 g/dm³ for mean 25.00 (6). Rinse pipette with B/burette with A, remove funnel, eye-level meniscus, swirl, dropwise near endpoint; any two (2). Propagate candidate's valid mean.

2. (12 marks) NaOH pale blue precipitate insoluble in excess; $Cu^{2}+$ consistent (4). Ammonia pale blue precipitate dissolves in excess giving deep-blue solution; $Cu^{2}+$ confirmed (4). White precipitate; sulfate; $Ba^{2}++SO4^{2}- \rightarrow BaSO4(s)$ (4). Do not claim tests actually carried out in simulation.

3. (12 marks) Rates .0400,.0323,.0238,.0159 s^-1 (4). Correct labelled axes, suitable scales, points/line (3). More concentrated thiosulfate: more reacting particles per volume, more frequent effective collisions, faster sulfur formation (3). Constant acid volume/concentration, total reaction volume, temperature, cross/observer/container, any two (2).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf